



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : SeismoSafe

Group No : 05, CSE370 Lab Section : 10, Spring 2026		
ID	Name	Contribution
24101481	Md Sahin Alam	EER diagram,Schema Diagram,Front End, BackEnd
24101380	Md. Rafid Rial	EER diagram,Schema Diagram,Front End, BackEnd
24101341	Imtiaz Ahmed Shovon	(Digitalized)EER and Schema Diagram,Front End, BackEnd

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Introduction

[Briefly describe the idea of your project]

Project Idea: SeismoSafe (Earthquake Vulnerable Building Detection & Risk Analysis System for Bangladesh)

Bangladesh is located near an active tectonic plate. Cities like Dhaka with high population density, unplanned buildings and weak constructional practices are under high risk of damage in earthquakes. Many structures are not built with proper safety guidelines. This increases the risk of damage during an Earthquake. . This creates a strong need for a smart system to monitor building safety and raise awareness.

SeismoSafe is a web-based system that identifies the vulnerable buildings and manages important safety information in one platform. It improves building safety and earthquake awareness in Bangladesh.

The system has four users. They are Admin, Owner, Inspector, and Tenant. They have separate dashboards. Owners can register the building with its respective information. Based on the information and emergency facilities it calculates and defines buildings risk score and safety level.

Tenants can submit complaints about safety issues with description and photo. The admin manages these complaints and assigns them to inspectors. Inspectors review the issues, update the status (Pending, Working, Solved), and give feedback with text and images.

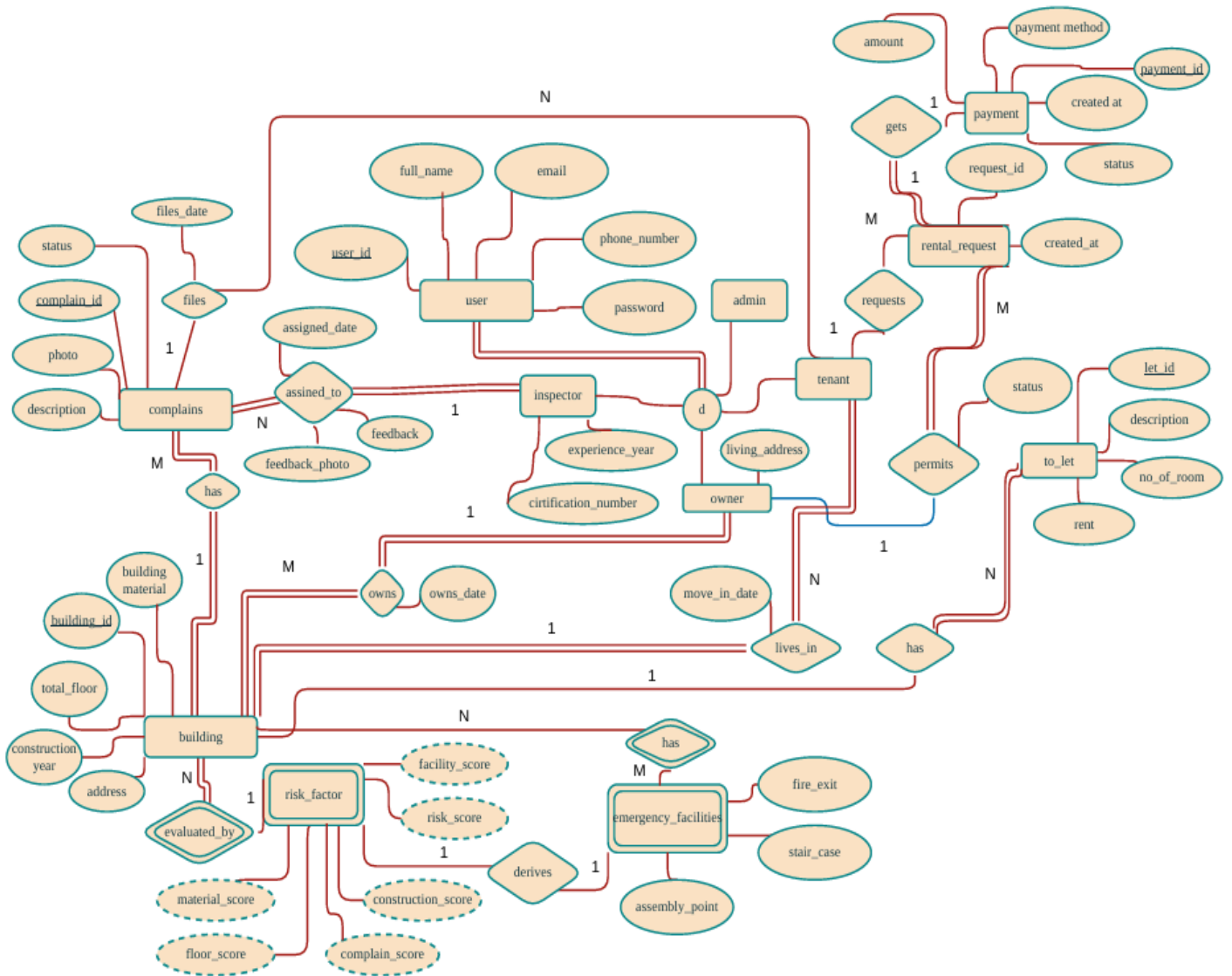
The admin has the power to control everything, including buildings, users, complaints, and overall statistics. This system also has a to-let and rental system where owners can post listings and tenants can request rentals. A secure payment system using SSLCommerz is integrated for easy transactions.

Overall, SeismoSafe is a website that helps to improve building safety, manage rentals, and increase earthquake awareness through a simple and centralized digital platform.

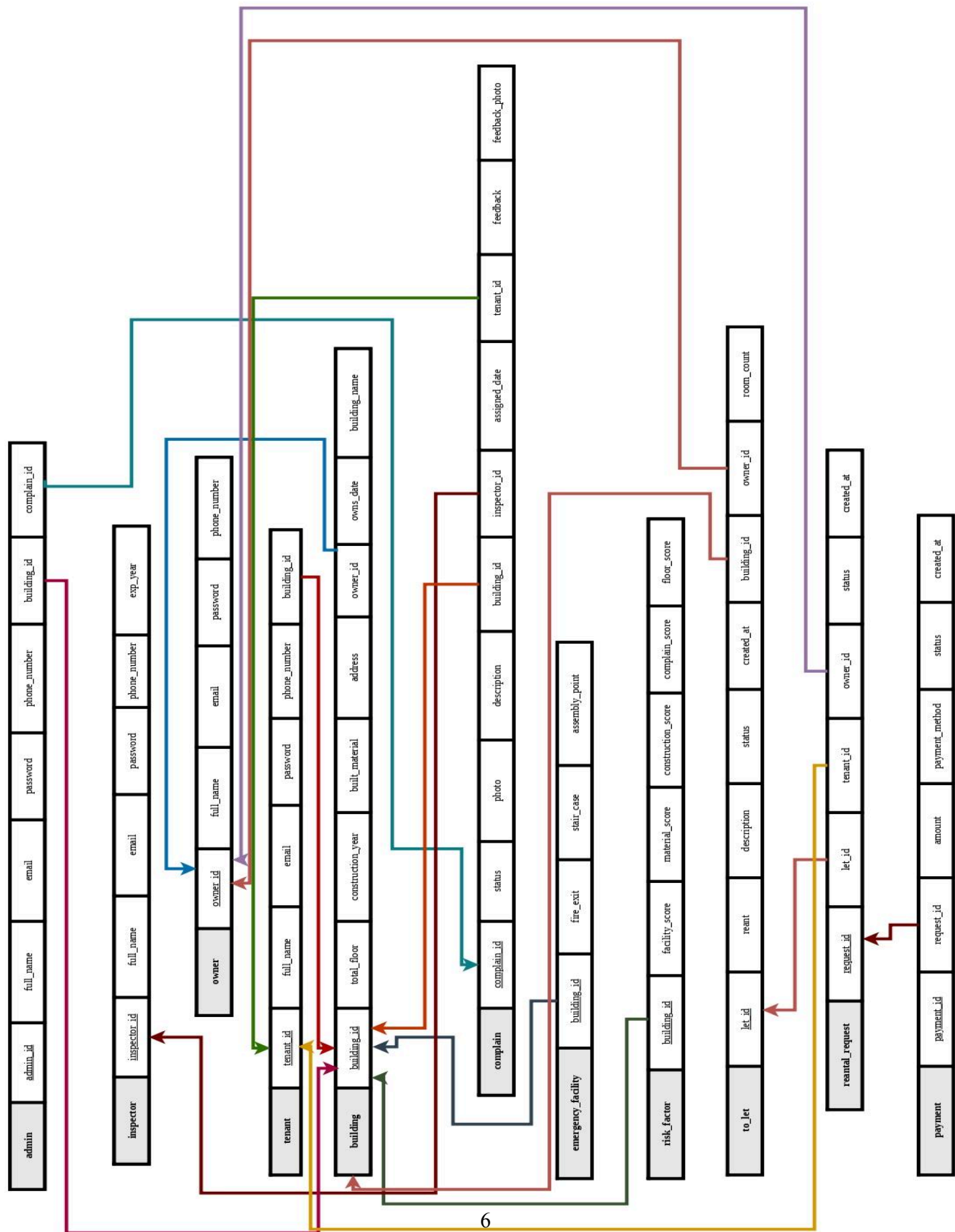
Project Features

ID, Name	Features [3 per member]	
24101481, Md Sahin Alam	Ft 1	Measure Vulnerability using Risk Score
	Ft 2	Admin Control Panel
	Ft 3	Inspector Control Panel
24101380, Md Rafid Rial	Ft 1	Complain Management
	Ft 2	Building Registry and own building Info Display
	Ft 3	To-let System
24101341, Imtiaz Ahmed Shovon	Ft 1	Profile Management
	Ft 2	Payment Gateway
	Ft 3	Role Base Access Control

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Ans:

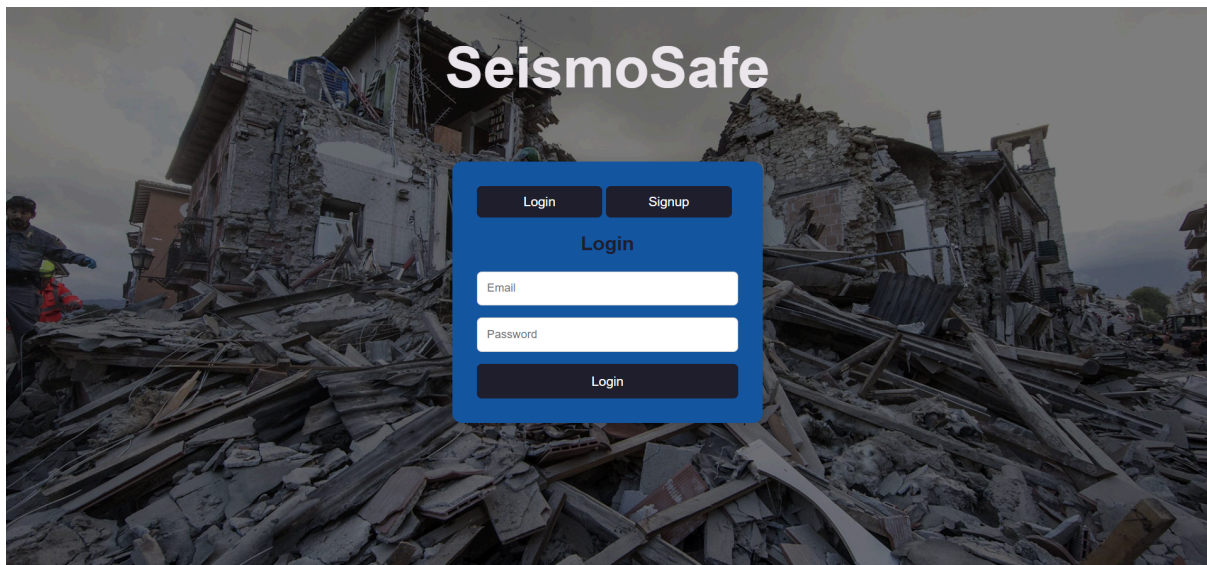
- a. **The converted Schema is already in 1NF as there are no multivalued, composite attributes or nested relations.**
- b. **The converted Schema is already in 2NF as there are no partial dependencies in any entity.**
- c. **The converted Schema is already in 3NF because there are no transitive dependencies present in any of the entities.**

Frontend Development

Briefly discuss about Frontend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 24101481, Name : Md Sahin Alam

Login Sign Up:



Admin Control Panel:

The image shows the 'Admin Dashboard' of the SeismoSafe application. On the left is a dark sidebar with the title 'Admin Panel' and a list of menu items: 'Add Building', 'Buildings', 'Add Inspector', 'Inspectors', 'Complaints', and 'Logout'. The main content area has a light blue header with the title 'Admin Dashboard' and a sub-header 'Add Building'. Below this is a form with several input fields: 'Building Name', 'Address', 'Select Owner' (a dropdown menu), 'Total Floor', 'Year', 'Material' (a dropdown menu), and a date field labeled 'mm/dd/yyyy'. At the bottom of the form is a dark button labeled 'Add Building'. Below the form is a section titled 'Buildings'.

Admin Panel

Add Building

Buildings

Add Inspector

Inspectors

Complaints

Logout

Buildings

ID	Name	Address	Owner	Floor	Year	Material	Date	Action
1	Mina Garden	M-53, West Merul Badda, Dhaka	11111	6	1990	Brick	1990-07-12	Delete
2	Shirin Tower	Merul,Dhaka	11111	8	1971	Mixed	2004-07-06	Delete
4	Brac	Dhaka	11111	15	2024	Concrete	2024-07-01	Delete
5	nikunjo	Savar,Dhaka	11112	6	1991	Brick	2021-04-04	Delete
6	Dld0	dgfhj	11111	5	1980	Concrete	2026-04-08	Delete
7	Abc	Dhaka	11111	6	1990	Mixed	2026-04-08	Delete
8	dfcgvhbj	Rajshahi	11112	4	1991	Brick	2026-04-16	Delete
9	SD	Dhaka	11112	5	1999	Steel Structure	2026-05-26	Delete
12	kkp	brac	11111	2	2010	Mixed	2026-05-22	Delete
13	Bangla	Gulshan 1,Dhaka	11112	3	2020	Mixed	2019-06-06	Delete

Total Buildings: 10

Add Inspector

Admin Panel

Add Building

Buildings

Add Inspector

Inspectors

Complaints

Logout

Add Inspector

Name

Email

Phone

Experience

Certification

Password

Add Inspector

Inspectors

ID	Name	Email	Phone	Experience
101	Inspector	i@gmail.com	3456789	3
102	sahil	sahil@gmail.com	09865433	3
103	ashik	ashikslm01976@gmail.com	01753774104	4

Admin Panel

Add Building

Buildings

Add Inspector

Inspectors

Complaints

Logout

Complaints

ID	Description	Status	Inspector	Date	Assign	Feedback
1	A crack has appeared on the building wall near the staircase and seems to be increasing gradually. The crack is visible both inside and outside the structure, causing safety concerns. Immediate inspection and repair are requested to prevent further damage.	Working	101	2026-04-16	Select Inspector mm / dd / yyyy Assign	the building has so many crack
9	I want to complain this building has so many crack	Working	101	2026-05-14	Select Inspector mm / dd / yyyy Assign	dfghnm
10	dfghjk	Solved	101	2026-05-06	Select Inspector mm / dd / yyyy Assign	ksjdkjskd
11	hbjhknkkjnhvbbjh hghhjjknkjh	Working	101	2026-05-23	Select Inspector mm / dd / yyyy Assign	No feedback yet

Admin Panel

Add Building

Buildings

Add Inspector

Inspectors

Complaints

Logout

9

I want to complain this building has so many crack

Working

101

2026-05-14

mm/dd/yyyy

dfghnm

Assign

10

dfghjk

Solved

101

2026-05-06

Select Inspector

mm/dd/yyyy

ksjdkjskd

Assign

11

hbjhknkikhvbbjh hjghjknkjh

Working

101

2026-05-23

Select Inspector

mm/dd/yyyy

No feedback yet

Assign

Total Complain: 4

Solved Complain: 1

Working Complain: 3

Pending Complain: 0

Inspector Control Panel:

Inspector Panel

My Complaints

Logout

My Assigned Complaints

ID	Tenant	Email	Phone	Building	Address	Complain	Photo	Status	Feedback	Inspector Photo	Update
1	tenant	t@gmail.com	1234563456	Mina Garden	M-53, West Merul Badda, Dhaka	A crack has appeared on the building wall near the staircase and seems to be increasing gradually. The crack is visible both inside and outside the structure, causing safety concerns. Immediate inspection and repair are requested to prevent further damage.		Working	the building has so many crack		Select Write feedback... Choose File No file chosen Update
9	tenant	t@gmail.com	1234563456	Mina Garden	M-53, West Merul Badda, Dhaka	I want to complain this building has so many crack		Working	dfghnm		Select Write feedback... Choose File No file chosen Update

Measure Vulnerability using RiskFactor:

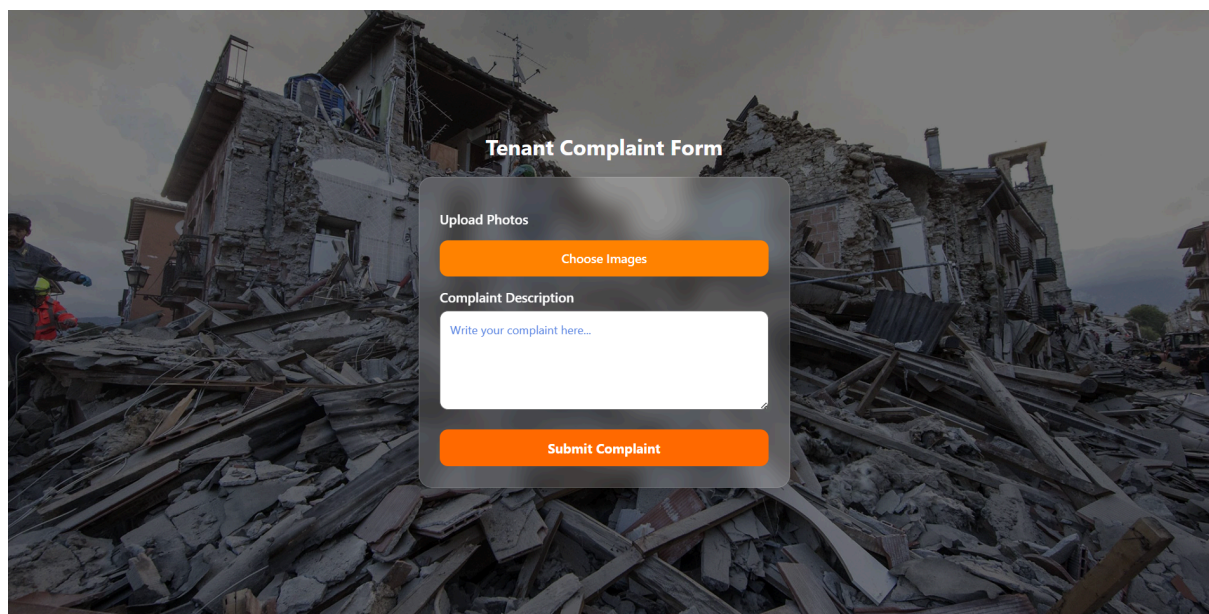
Building Risk Report

Search by Building Name

Search

Building ID	Building Name	Owner Name	Risk Score	Risk Level
1	Mina Garden	owner	0.60	MEDIUM
2	Shirin Tower	owner	0.62	HIGH
4	Brac	owner	0.30	LOW
5	nikunjo	owner2	0.50	MEDIUM
6	Did0	owner	0.36	MEDIUM
7	Abc	owner	0.42	MEDIUM
8	dfcgvhbj	owner2	0.50	MEDIUM
9	SD	owner2	0.32	MEDIUM
12	kkp	owner	0.26	LOW
13	Bangla	owner2	0.20	LOW

Contribution of ID : 20101380, Name : Md. Rafid Rial
Tenant HomePage(Complain section and My Complain)



[← Back](#)

My Complains

ID	Building	Description	Status	Photo
1	Mina Garden	A crack has appeared on the building wall near the staircase and seems to be increasing gradually. The crack is visible both inside and outside the structure, causing safety concerns. Immediate inspection and repair are requested to prevent further damage.	Working	View
9	Mina Garden	I want to comlain this building has so many crack	Working	View
10	Shirin Tower	dfghjk	Solved	View
11	Shirin Tower	hbjhknkkjhvbvjh hjghjjknkj	Working	View

To-Let System:

Post To-Let

Post To-Let

Available To-Let Listings

Shirin Tower

Owner: owner

Rooms: 5

Rent: 30000.00 BDT

2nd Floor, 2 Master bedroom, 1 bedroom, 1 drawing Room, with balcony in all Rooms

Status: [Available](#)

Requested

SD

Owner: owner2

Rooms: 3

Rent: 25000.00 BDT

To BedRoom and One guest Room

Status: [Available](#)

Requested

My Rental Requests

Building: Shirin Tower

Owner: owner

Rent: 30000.00

Status: Accepted

[Pay Now](#)

Building: SD

Owner: owner2

Rent: 25000.00

Status: Accepted

[Pay Now](#)

Tenant Requests

Tenant: tenant

Building: SD

Rent: 25000.00

Rooms: 3

Status: Accepted

[Accept](#)

[Reject](#)

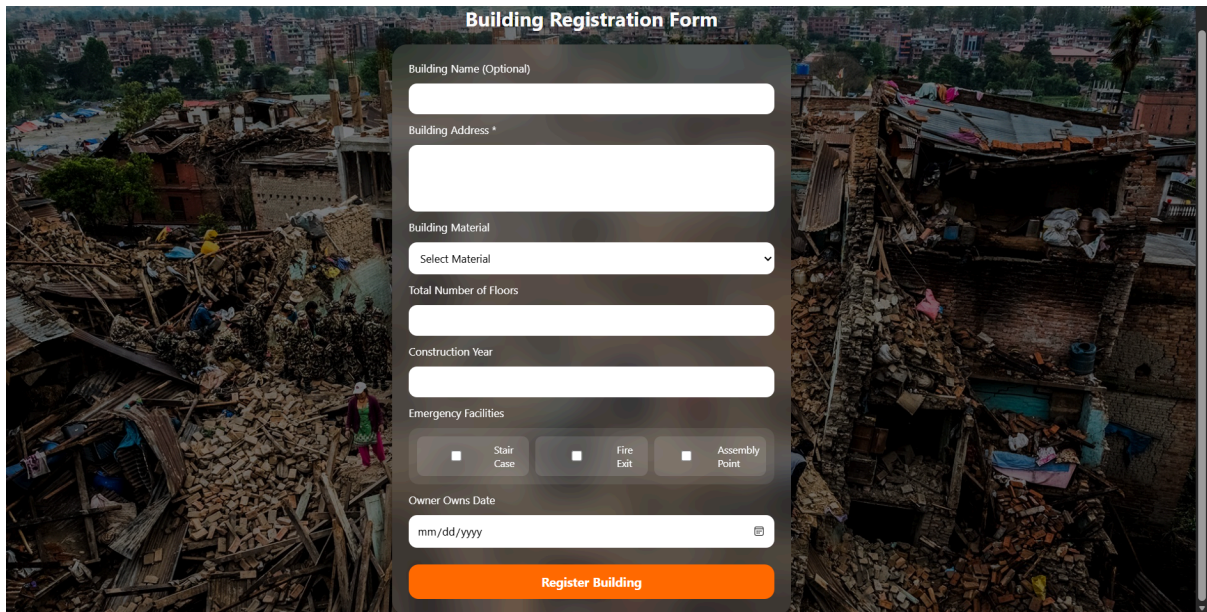
OwnerHomePage(Building Resigstry and Info):

SeismoSafe

HOMEREGISTRATIONMy BUILDINGSRiskScorePost To-LetREQUESTSLOGOUTowner2

"Be self-aware and make others aware."

An earthquake is the sudden shaking of the Earth's surface caused by the release of energy from inside the Earth's crust. It mainly occurs due to the movement of tectonic plates along faults. The point where it starts is called the focus, and the point directly above it on the surface is the epicenter. Earthquakes are measured using a seismograph and expressed in magnitude. They can cause serious damage such as building collapse, landslides, and sometimes tsunamis. Although earthquakes cannot be prevented, proper preparedness, strong construction, and awareness can reduce damage and save lives during such natural disasters.



Building Registration Form

Building Name (Optional)

Building Address *

Building Material

Total Number of Floors

Construction Year

Emergency Facilities
☐ Stair Case ☐ Fire Exit ☐ Assembly Point

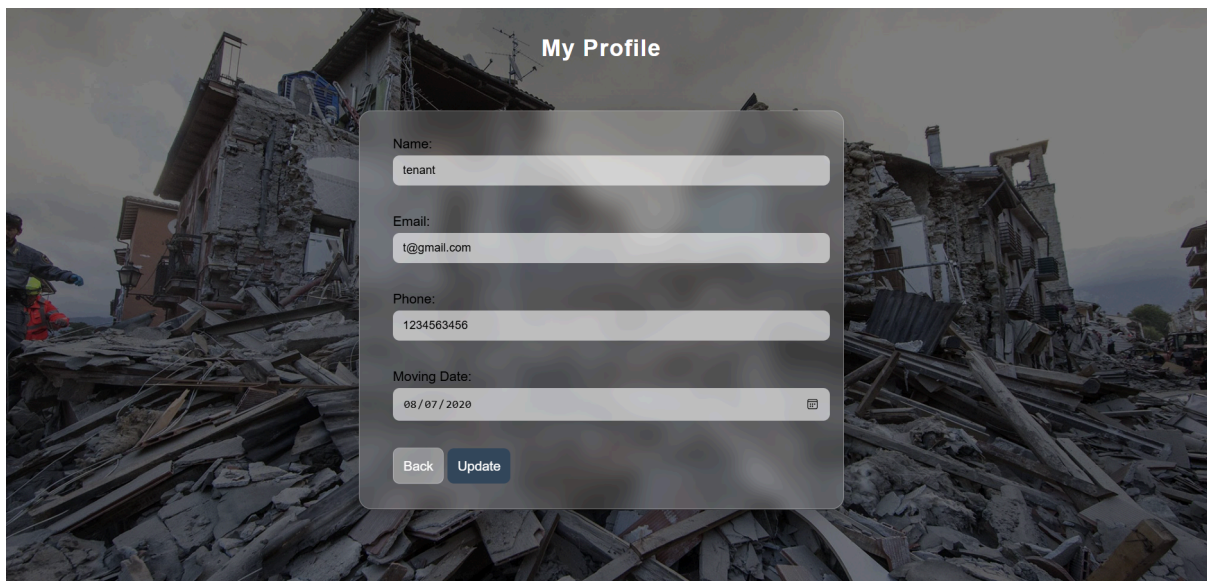
Owner Owns Date

Register Building

My Buildings with Risk Score							
ID	Name	Address	Floors	Year	Material	Risk Score	Status
5	nikunjo	Savar,Dhaka	6	1991	Brick	0.50	MEDIUM
8	dfcgvhbj	Rajshahi	4	1991	Brick	0.50	MEDIUM
9	SD	Dhaka	5	1999	Steel Structure	0.32	MEDIUM
13	Bangla	Gulshan 1,Dhaka	3	2020	Mixed	0.20	LOW

Contribution of ID : 20101341, Name : Imtiaz Ahmed Shovon

Profile Management:



My Profile

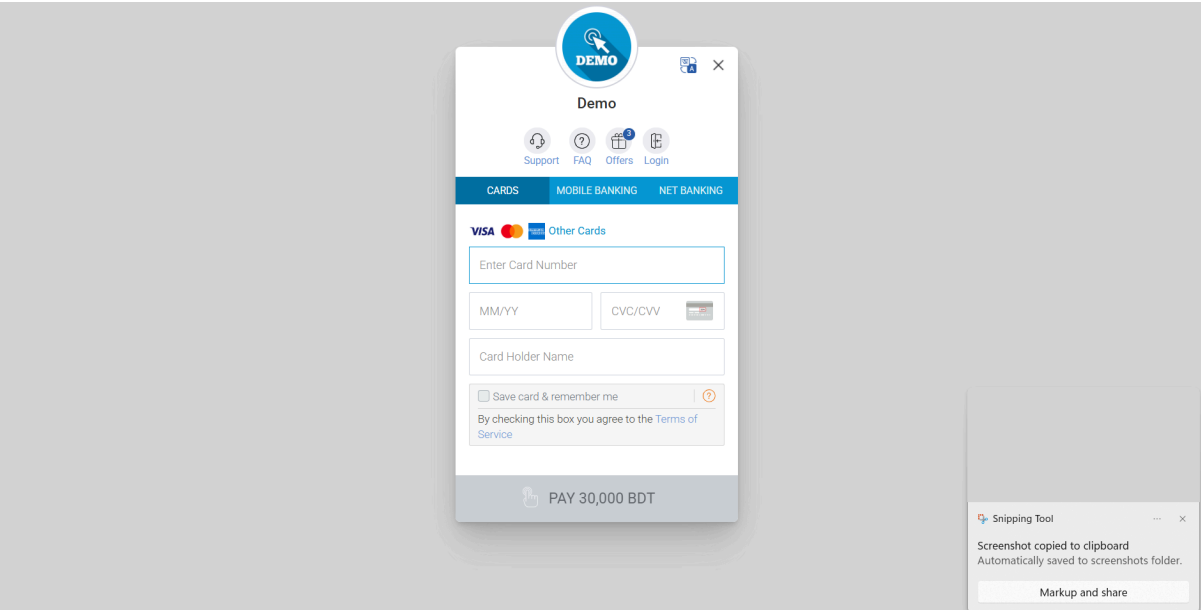
Name:

Email:

Phone:

Moving Date:

Back Update



Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

Contribution of ID : 24101481, Name : Md Sahin Alam

LogIn Sign Up:

1.Measure Vulnerability using Risk score :

A dedicated view where users can see relevant information related to all the buildings with risk score to understand the vulnerability of the buildings. Users can also search for a specific building.

```
<?php
$conn = mysqli_connect("localhost", "root", "",
"project_370");

if (!$conn) {
    die("DB Connection Failed");
}

$search = trim($_GET['search'] ?? '');

if ($search == '') {

    $sql = "SELECT
            b.*,
            o.full_name,
            COUNT(c.complain_id) AS total_complain
        FROM building b
        JOIN owner o ON b.owner_id = o.owner_id
        LEFT JOIN complain c ON b.building_id =
c.building_id
        GROUP BY b.building_id";

} else {
```

```

$sql = "SELECT
        b.*,
        o.full_name,
        COUNT(c.complain_id) AS total_complain
FROM building b
JOIN owner o ON b.owner_id = o.owner_id
LEFT JOIN complain c ON b.building_id =
c.building_id
WHERE b.building_name LIKE '%$search%'
GROUP BY b.building_id";
}

$result = mysqli_query($conn, $sql);

if (!$result) {
    die("Query Error: " . mysqli_error($conn));
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Risk Report</title>

    <style>
        body {
            font-family: Arial;
            background: #f4f6f9;
            padding: 20px;
        }

        form {
            margin-bottom: 15px;

```

```

}

input[type="text"] {
    padding: 10px;
    width: 250px;
    border-radius: 5px;
    border: 1px solid #ccc;
}

button {
    padding: 10px 15px;
    background: #1e1e2f;
    color: white;
    border: none;
    border-radius: 5px;
    cursor: pointer;
}

button:hover {
    background: #34345a;
}

table {
    width: 100%;
    border-collapse: collapse;
    background: white;
}

th, td {
    padding: 12px;
    border: 1px solid #ddd;
    text-align: center;
}

```

```

        th {
            background: #1e1e2f;
            color: white;
        }
    </style>
</head>

<body>

<h2>Building Risk Report</h2>

<!-- search-->
<form method="GET" id="searchForm">
    <input type="text"
        name="search"
        id="searchInput"
        value="<?= $search ?>"
        placeholder="Search by Building Name">

    <button type="submit">Search</button>
</form>

<table>
<tr>
    <th>Building ID</th>
    <th>Building Name</th>

    <th>Owner Name</th>
    <th>Risk Score</th>
    <th>Risk Level</th>
</tr>

<?php
while ($row = mysqli_fetch_assoc($result)) {

```

```

$fire_exit = $row['fire_exit'] ?? 0;
$stair_case = $row['stair_case'] ?? 0;
$assembly_point = $row['assembly_point'] ?? 0;
$complain = $row['total_complain'] ?? 0;

$facility_count = 0;
if($fire_exit) $facility_count++;
if($stair_case) $facility_count++;
if($assembly_point) $facility_count++;

if($facility_count == 3) $facility_score = 0;
elseif($facility_count == 2) $facility_score = 0.4;
elseif($facility_count == 1) $facility_score = 0.7;
else $facility_score = 0;

$material = strtolower(trim($row['built_material']
?? ''));

if($material == "steel structure") $material_score
=1;
elseif($material == "brick") $material_score = 0.6;
elseif($material == "mixed") $material_score = 0.3;
elseif($material == "concrete") $material_score =
0.1;
else $material_score = 0.5;

$year = $row['construction_year'] ?? 0;

if($year > 2015) $construction_score = 0.2;
elseif($year >= 2000) $construction_score = 0.5;
else $construction_score = 1;

if($complain == 0) $complain_score = 0;

```



```

elseif($complain <= 3) $complain_score = 0.5;
else $complain_score = 1;

$floor = $row['total_floor'] ?? 0;

if($floor <= 3) $floor_score = 0.2;
elseif($floor <= 7) $floor_score = 0.5;
else $floor_score = 1;

$risk_score = ($facility_score + $material_score +
$construction_score + $floor_score + $complain_score) /
5;

if($risk_score <= 0.3) $status = "LOW";
elseif($risk_score <= 0.6) $status = "MEDIUM";
else $status = "HIGH";
?>

<tr>
    <td><?= $row['building_id'] ?></td>
    <td><?= $row['building_name'] ?></td>
    <td><?= $row['full_name'] ?></td>
    <td><?= number_format($risk_score, 2) ?></td>
    <td><?= $status ?></td>
</tr>

<?php } ?>

</table>

<script>
document.getElementById("searchInput").addEventListener
("input", function () {

```

```

        if (this.value.trim() === "") {
            window.location = "risk.php";
        }
    });
</script>

</body>
</html>

```

2.Admin Control Panel :

Admin has full control over the system. Admin can add and delete buildings, monitor complaints, assign inspectors, and monitor inspector feedbacks, also can see how many complain at what status and total building and complain. The Admin panel acts as the central management system for all activities.

```

<?php
session_start();

$conn = mysqli_connect("localhost", "root", "",
"project_370");

if (!$conn) {
    die("DB Connection Failed");
}

//Inspector Add
if (isset($_POST['add_inspector'])) {

    $full_name = $_POST['full_name'];
    $email = $_POST['email'];
    $phone_number = $_POST['phone_number'];
    $exp_year = $_POST['exp_year'];
    $certification_num = $_POST['certification_num'];
    $password = $_POST['password'];

    mysqli_query($conn, "INSERT INTO inspector

```

```

        (full_name, email, phone_number, exp_year,
certification_num, password)
    VALUES
        ('$full_name', '$email', '$phone_number',
'$exp_year', '$certification_num', '$password')");
}

//Building Add
if (isset($_POST['add_building'])) {

    $building_name = $_POST['building_name'];
    $address = $_POST['address'];
    $owner_id = $_POST['owner_id'];
    $total_floor = $_POST['total_floor'];
    $construction_year = $_POST['construction_year'];
    $built_material = $_POST['built_material'];
    $owns_date = $_POST['owns_date'];

    mysqli_query($conn, "INSERT INTO building
        (building_name, address, owner_id, total_floor,
construction_year, built_material, owns_date)
    VALUES
        ('$building_name', '$address', '$owner_id',
'$total_floor', '$construction_year',
'$built_material', '$owns_date')");
}

//Delete Building
if (isset($_GET['delete'])) {
    $id = $_GET['delete'];
    mysqli_query($conn, "DELETE FROM building WHERE
building_id=$id");
}

```

```

//Assigned Inspector to a complain
if (isset($_POST['assign_inspector'])) {

    $cid = $_POST['complain_id'];
    $iid = $_POST['inspector_id'];
    $assigned_date = $_POST['assigned_date'];

    mysqli_query($conn, "UPDATE complain
        SET inspector_id='$iid',
            assigned_date='$assigned_date'
        WHERE complain_id='$cid'");
}

//For display all data
$buildings = mysqli_query($conn, "SELECT * FROM
building order by building_name asc");
$complains = mysqli_query($conn, "SELECT * FROM
complain");
$inspectors = mysqli_query($conn, "SELECT * FROM
inspector");
$owners = mysqli_query($conn, "SELECT owner_id FROM
owner");

$total_buildings =
mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*)
AS total FROM building"))['total'];
$total_complaints =
mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*)
AS total FROM complain"))['total'];
$solved_complaints =
mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*)
AS total FROM complain WHERE
status='Solved'"))['total'];

```

```

$working_complains =
mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*)
AS total FROM complain WHERE
status='Working'"))['total'];
$pending_complains =
mysqli_fetch_assoc(mysqli_query($conn, "SELECT COUNT(*)
AS total FROM complain WHERE
status='Pending'"))['total'];
?>

<!DOCTYPE html>
<html>
<head>
    <title>Admin Dashboard</title>
    <link rel="stylesheet" href="adminHomePage.css">
</head>

<body>

<div class="sidebar">
    <h2>Admin Panel</h2>
    <a href="#add">Add Building</a>
    <a href="#building">Buildings</a>
    <a href="#addInsp">Add Inspector</a>
    <a href="#inspectors">Inspectors</a>
    <a href="#complain">Complaints</a>
    <a href="logOut.php">Logout</a>
</div>

<div class="main">

<h1>Admin Dashboard</h1>

```

```

<section id="add">
<h2>Add Building</h2>

<form method="POST">
<input type="text" name="building_name"
placeholder="Building Name" required>
<input type="text" name="address" placeholder="Address"
required>

<select name="owner_id" required>
<option value="">Select Owner</option>
<?php while($o = mysqli_fetch_assoc($owners)) { ?>
<option value="<?= $o['owner_id'] ?>"><?=
$o['owner_id'] ?></option>
<?php } ?>
</select>

<input type="number" name="total_floor"
placeholder="Total Floor" required>
<input type="number" name="construction_year"
placeholder="Year" required>

<select name="built_material" required>
<option value="">Material</option>
<option>Brick</option>
<option>Concrete</option>
<option>Steel</option>
<option>Mixed</option>
</select>

<input type="date" name="owns_date" required>

<button name="add_building">Add Building</button>
</form>

```

```

</section>

<hr>

<section id="building">
<h2>Buildings</h2>

<table>
<tr>
<th>ID</th><th>Name</th><th>Address</th><th>Owner</th><
th>Floor</th><th>Year</th><th>Material</th><th>Date</th>
><th>Action</th>
</tr>

<?php while($b = mysqli_fetch_assoc($buildings)) { ?>
<tr>
<td><?= $b['building_id'] ?></td>
<td><?= $b['building_name'] ?></td>
<td><?= $b['address'] ?></td>
<td><?= $b['owner_id'] ?></td>
<td><?= $b['total_floor'] ?></td>
<td><?= $b['construction_year'] ?></td>
<td><?= $b['built_material'] ?></td>
<td><?= $b['owns_date'] ?></td>
<td><a href="?delete=<?= $b['building_id']
?>">Delete</a></td>
</tr>
<?php } ?>
</table>

<div class="summary-box">
Total Buildings: <?= $total_buildings ?>
</div>

```

```

</section>

<hr>

<section id="addInsp">
<h2>Add Inspector</h2>

<form method="POST">
<input type="text" name="full_name" placeholder="Name"
required>
<input type="email" name="email" placeholder="Email"
required>
<input type="text" name="phone_number"
placeholder="Phone" required>
<input type="number" name="exp_year"
placeholder="Experience" required>
<input type="text" name="certification_num"
placeholder="Certification" required>
<input type="password" name="password"
placeholder="Password" required>

<button name="add_inspector">Add Inspector</button>
</form>
</section>

<hr>

<section id="inspectors">
<h2>Inspectors</h2>

<table>

```



```

<tr>
<th>ID</th><th>Name</th><th>Email</th><th>Phone</th><th>
>Experience</th>
</tr>

<?php
$insp = mysqli_query($conn, "SELECT * FROM inspector");
while($i = mysqli_fetch_assoc($insp)) {
?>
<tr>
<td><?= $i['inspector_id'] ?></td>
<td><?= $i['full_name'] ?></td>
<td><?= $i['email'] ?></td>
<td><?= $i['phone_number'] ?></td>
<td><?= $i['exp_year'] ?></td>
</tr>
<?php } ?>

</table>
</section>

<hr>

<section id="complain">
<h2>Complaints</h2>

<table>
<tr>
<th>ID</th>
<th>Description</th>
<th>Status</th>
<th>Inspector</th>
<th>Date</th>

```

```

<th>Assign</th>
<th>Feedback</th>
</tr>

<?php while($c = mysqli_fetch_assoc($complains)) { ?>
<tr>
<td><?= $c['complain_id'] ?></td>
<td><?= $c['description'] ?></td>
<td><?= $c['status'] ?></td>
<td><?= $c['inspector_id'] ?></td>
<td><?= $c['assigned_date'] ?></td>

<td>
<form method="POST">
<input type="hidden" name="complain_id" value="<?=
$c['complain_id'] ?>">

<select name="inspector_id" required>
<option value="">Select Inspector</option>

<?php
mysqli_data_seek($inspectors, 0);
while($i = mysqli_fetch_assoc($inspectors)) {
?>
<option value="<?= $i['inspector_id'] ?>">
<?= $i['inspector_id'] ?>
</option>
<?php } ?>
</select>

<input type="date" name="assigned_date" required>

<button name="assign_inspector">Assign</button>
</form>

```

```

</td>
<td><?= $c['feedback'] ?? 'No feedback yet' ?></td>
</tr>
<?php } ?>

</table>

<div class="summary-box">Total Complains: <?=
$total_complaints ?></div>
<div class="summary-box">Solved Complains: <?=
$solved_complains ?></div>
<div class="summary-box">Working Complains: <?=
$working_complains ?></div>
<div class="summary-box">Pending Complains: <?=
$pending_complains ?></div>

</section>

</div>

</body>
</html>

```

3. Inspector Control Panel:

The inspector can see his assigned works on his panel. He can update the progress of his investigation on a complain and give feedback on it using text and photo.

```

<?php
session_start();

$conn = mysqli_connect("localhost", "root", "",
"project_370");

if (!$conn) {

```

```

        die("Database connection failed: " .
mysqli_connect_error());
    }

    if (!isset($_SESSION['inspector_id'])) {
        header("Location: logInSignUp.html");
        exit();
    }

    $inspector_id = $_SESSION['inspector_id'];

    //Update status with Feedback
    if (isset($_POST['update_status'])) {

        $complain_id = $_POST['complain_id'];
        $status = $_POST['status'];
        $feedback = $_POST['feedback'];

        $photo_name = "";

        //feedBack Photo uplaod
        if (!empty($_FILES['feedback_photo']['name'])) {

            $photo_name = time() . "_" .
$_FILES['feedback_photo']['name'];
            $tmp = $_FILES['feedback_photo']['tmp_name'];

            move_uploaded_file($tmp, "uploads/" .
$photo_name);
        }

        $sql = "UPDATE complain
                SET status='$status',
                feedback='$feedback'";
    }

```

```

    if ($photo_name != "") {
        $sql .= ", feedback_photo='$photo_name'";
    }

    $sql .= " WHERE complain_id='$complain_id'
            AND inspector_id=$inspector_id";

    mysqli_query($conn, $sql);
}

//Complain Table
$sql = "SELECT
        c.complain_id,
        c.description,
        c.photo,
        c.status,
        c.feedback,
        c.feedback_photo,
        t.full_name,
        t.email,
        t.phone_number,
        b.building_name,
        b.address
    FROM complain c
    JOIN tenant t ON c.tenant_id = t.tenant_id
    JOIN building b ON c.building_id =
b.building_id
    WHERE c.inspector_id = $inspector_id";

$result = mysqli_query($conn, $sql);

if (!$result) {
    die("Query failed: " . mysqli_error($conn));
}

```

```

}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Inspector Dashboard</title>
    <link rel="stylesheet"
href="inspectorHomePage.css">

    <style>
        textarea {
            width: 100%;
            height: 60px;
            margin-top: 5px;
        }

        .thumb {
            width: 60px;
            border-radius: 5px;
        }

        .feedback-img {
            width: 60px;
            border-radius: 5px;
        }
    </style>
</head>

<body>

<div class="sidebar">
    <h2>Inspector Panel</h2>
    <a href="#">My Complaints</a>

```

```

    <a href="logOut.php">Logout</a>
</div>

<div class="main">
    <h1 class="name">My Assigned Complaints</h1>

    <table border="1" cellpadding="8">
        <tr>
            <th>ID</th>
            <th>Tenant</th>
            <th>Email</th>
            <th>Phone</th>
            <th>Building</th>
            <th>Address</th>
            <th>Complain</th>
            <th>Photo</th>
            <th>Status</th>
            <th>Feedback</th>
            <th>feedBack Photo</th>
            <th>Update</th>
        </tr>

        <?php
        if (mysqli_num_rows($result) > 0) {
            while ($row = mysqli_fetch_assoc($result))
        {
            ?>
            <tr>
                <td><?= $row['complain_id'] ?></td>
                <td><?= $row['full_name'] ?></td>
                <td><?= $row['email'] ?></td>
                <td><?= $row['phone_number'] ?></td>
                <td><?= $row['building_name'] ?></td>
                <td><?= $row['address'] ?></td>

```

```

        <td><?= $row['description'] ?></td>

        <td>
            <a href="uploads/<?= $row['photo'] ?>"
target="_blank">
                
            </a>
        </td>

        <td><?= $row['status'] ?></td>

        <td><?= $row['feedback'] ?? 'No feedback'
?></td>

        <td>
            <?php
if(!empty($row['feedback_photo'])) { ?>
                
            <?php } else { echo "No Image"; } ?>
        </td>

        <td>
            <form method="POST"
enctype="multipart/form-data">

                <input type="hidden"
name="complain_id" value="<?= $row['complain_id'] ?>">

                <select name="status" required>
                    <option
value="">Select</option>

```



```

                <option
value="Pending">Pending</option>
                <option
value="Working">Working</option>
                <option
value="Solved">Solved</option>
            </select>

            <textarea name="feedback"
placeholder="Write feedback..." required></textarea>

            <input type="file"
name="feedback_photo">

            <button type="submit"
name="update_status">Update</button>
        </form>
    </td>

</tr>
<?php
    }
    } else {
        echo "<tr><td colspan='12'>No complaints
assigned yet</td></tr>";
    }
?>

</table>
</div>

</body>
</html>

```

4.Login SignUp:

Different User can sign Up According to their Role and after sign in they will get different dashboard according to their role.

Contribution of ID : 20101380, Name : Md. Rafid Rial

1.Building Registry and own Building Info Display:

Owners can register buildings by filling out a registration form. Building details such as building name, address, total floors, construction year, material type, and emergency facilities are stored in the database. The owner can also see his own building details.

2.Complain Management:

Tenants can submit complains about building problems such as cracks or safety issues. Complain is stored in the database and can be assigned to inspectors by the admin. Complaint status can be updated as Pending, working or Solved by the Inspector.

3.To-let Management System:

The owner can give a tolet post on the tenant's side, can view the tenant's request and accept it. Tenant can request for rent and also can pay the rental fee.

```
<?php
session_start();

$conn = mysqli_connect("localhost", "root", "", "project_370");

if (!$conn) {
    die("DB Connection Failed");
}

/* ===== CHECK OWNER LOGIN ===== */
if (!isset($_SESSION['owner_id'])) {
    header("Location: login.php");
    exit();
}

$owner_id = $_SESSION['owner_id'];

/* ===== INSERT TO LET ===== */
$message = "";
```

```

if (isset($_POST['submit'])) {

    $building_id = $_POST['building_id'];
    $room_count   = $_POST['room_count'];
    $rent         = $_POST['rent'];
    $description   = $_POST['description'];

    $sql = "INSERT INTO to_let
            (building_id, owner_id, room_count, rent, description)
            VALUES
            ('$building_id', '$owner_id', '$room_count', '$rent',
'$description')";

    if (mysqli_query($conn, $sql)) {
        $message = "✅ To-Let Posted Successfully!";
    } else {
        $message = "❌ Error: " . mysqli_error($conn);
    }
}

?>
<?php
session_start();

$conn = mysqli_connect("localhost", "root", "", "project_370");

if (!$conn) {
    die("DB Connection Failed");
}

/* ===== GET TENANT ===== */
$tenant_id = $_SESSION['tenant_id'] ?? 0;

/* ===== FETCH ONLY AVAILABLE ===== */
$sql = "SELECT
        t.let_id,
        t.owner_id,
        t.building_id,
        t.room_count,
        t.rent,
        t.description,
        t.status,
        b.building_name,
        o.full_name AS owner_name

```

```

        FROM to_let t
        JOIN building b ON t.building_id = b.building_id
        JOIN owner o ON t.owner_id = o.owner_id
        WHERE t.status = 'Available'
        ORDER BY t.let_id DESC";

$result = mysqli_query($conn, $sql);

if (!$result) {
    die("Query Error: " . mysqli_error($conn));
}
?>

```

```

<?php
session_start();

$conn = mysqli_connect("localhost", "root", "", "project_370");

if (!$conn) {
    die("DB Connection Failed: " . mysqli_connect_error());
}

/* ===== CHECK LOGIN ===== */
if (!isset($_SESSION['tenant_id'])) {
    die("You must login as tenant first.");
}

/* ===== GET DATA FROM FORM ===== */
$let_id = $_POST['let_id'] ?? 0;
$tenant_id = $_POST['tenant_id'] ?? 0;
$owner_id = $_POST['owner_id'] ?? 0;

/* ===== VALIDATION ===== */
if ($let_id == 0 || $tenant_id == 0 || $owner_id == 0) {
    die("Invalid request data.");
}

/* ===== CHECK DUPLICATE REQUEST ===== */
$check = "SELECT * FROM rental_request
        WHERE let_id='$let_id'
        AND tenant_id='$tenant_id'";

```

```

$checkResult = mysqli_query($conn, $check);

if (mysqli_num_rows($checkResult) > 0) {

    echo "<script>
        alert('You already sent a request for this property!');
        window.location.href='vacancy.php';
    </script>";

    exit();
}

/* ===== INSERT REQUEST ===== */
$sql = "INSERT INTO rental_request
        (let_id, tenant_id, owner_id, status)
        VALUES
        ('$let_id', '$tenant_id', '$owner_id', 'Pending')";

if (mysqli_query($conn, $sql)) {

    echo "<script>
        alert('Request Sent Successfully!');
        window.location.href='vacancy.php';
    </script>";

} else {
    echo "Error: " . mysqli_error($conn);
}
?>
<?php
session_start();

$conn = mysqli_connect("localhost", "root", "", "project_370");

if (!$conn) {
    die("DB Connection Failed");
}

/* ===== GET DATA ===== */
$request_id = $_POST['request_id'] ?? 0;
$action = $_POST['action'] ?? '';

if ($request_id == 0 || $action == '') {

```

```

        header("Location: tenantHomePage.php");
        exit();
    }

    /* ===== UPDATE REQUEST STATUS ===== */
    $sql = "UPDATE rental_request
            SET status = '$action'
            WHERE request_id = '$request_id'";

    if (mysqli_query($conn, $sql)) {

        echo "<script>
            alert('Request $action Successfully!');
            window.location.href='ownerHomePage.php';
        </script>";

    } else {
        echo "Error: " . mysqli_error($conn);
    }
?>

```

Contribution of ID : 20101341, Name : Imtiaz Ahmed Shovon

1.Profile Management:

Owner and tenant can update their information and manage their respective profile.

```

<?php
session_start();
include "config.php";

if(isset($_SESSION['owner_id'])) {
    $table = "owner";
    $id_col = "owner_id";
    $id = $_SESSION['owner_id'];
}

```

```

elseif(isset($_SESSION['tenant_id'])) {
    $table = "tenant";
    $id_col = "tenant_id";
    $id = $_SESSION['tenant_id'];
}
elseif(isset($_SESSION['inspector_id'])) {
    $table = "inspector";
    $id_col = "inspector_id";
    $id = $_SESSION['inspector_id'];
}
else{
    die("No user logged in");
}

$sql = "SELECT * FROM $table WHERE $id_col='$id'";
$result = mysqli_query($conn, $sql);
$user = mysqli_fetch_assoc($result);
?>

<!DOCTYPE html>
<html>
<head>
    <title>My Profile</title>

    <link rel="stylesheet" href="profile.css">
</head>

<body>

<h2>My Profile</h2>

<form action="update_profile.php" method="POST">

```

```

<input type="hidden" name="table" value="<?php echo
$table; ?>">
<input type="hidden" name="id_col" value="<?php echo
$id_col; ?>">
<input type="hidden" name="id" value="<?php echo
$user[$id_col]; ?>">

Name:<br>
<input type="text" name="name" value="<?php echo
$user['full_name']; ?>"><br><br>

Email:<br>
<input type="email" name="email" value="<?php echo
$user['email']; ?>"><br><br>

Phone:<br>
<input type="text" name="phone" value="<?php echo
$user['phone_number']; ?>"><br><br>

<?php if($table=="owner"){ ?>
Address:<br>
<input type="text" name="address" value="<?php echo
$user['present_address']; ?>"><br><br>
<a href="ownerHomePage.php" class="back-btn">Back</a>
<?php } ?>

<?php if($table=="tenant"){ ?>
Moving Date:<br>
<input type="date" name="moving_date" value="<?php echo
$user['moving_date']; ?>"><br><br>
<a href="tenantHomePage.php" class="back-btn">Back</a>
<?php } ?>
<button type="submit">Update</button>

```



```
</form>

</body>
</html>
```

2.Payment Gateway:

Integrated SSL commerce for doing rental payments by tenants or collecting rents by the owner.

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

<https://github.com/shovon05/flex370>

Conclusion

SeismoSafe is a complete and practical system designed to improve building safety and management in Bangladesh, especially in earthquake-prone areas. It brings together building registration, risk analysis, complaint handling, inspection, rental management, and payment processing in one platform.

By using a role-based system, it ensures that Admins, Owners, Inspectors, and Tenants can efficiently perform their tasks with clear responsibilities. The risk score feature helps identify vulnerable buildings, while the complaint and inspection system improves communication and quick problem resolution.

Overall, SeismoSafe aims to make building management smarter, safer, and more organized, while also increasing awareness about earthquake risks and ensuring better safety for residents.

References

W3Schools Online Web Tutorials

https://www.youtube.com/watch?v=at19OmH2Bg4&list=PLu0W_9lII9aik

XkRE0WxDt1vozo3hnmtR

https://www.youtube.com/watch?v=WlpnK2NMX_U&t=2s

https://youtu.be/CQZxcoQeo5c?si=Itw_FR0c9Hj6JHJO